RAJALAKSHMI ENGINEERING COLLEGE

B.E. COMPUTER SCIENCE AND ENGINEERING
(CYBER SECURITY)

CR23532 – CYBER AND DIGITAL FORENSICS

CASE STUDY ASSIGNMENTS – II
CONSOLIDATED REPORT

STUDENT INFORMATION	DETAILS
Student Name	Mebish V D
Register / Roll Number	241901053
Department	CSE (Cyber Security)
Subject Code	CR23532
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1. ASSIGNMENT OVERVIEW AND SCOPE

This report addresses all five case-study assignments specified in the CR23532 – Cyber and Digital Forensics Assignment-II brief. The scenarios are treated as hypothetical educational case studies. The report explains defensible workflows, likely evidence sources, interpretation methods, and limitations. It does not claim that a real device was seized, that a forensic tool was run, or that any real-world finding was established.

Assignment	Case-study title	Core focus	Required deliverables covered
1	Suspected Unauthorized Access to a College Server	Investigation phases, evidence preservation, timeline, legal/ethical issues	Flowchart, evidence table, case analysis, sample report structure
2	Linking a Suspicious File to a User Using File-System Evidence	System inventory, metadata, hidden/encrypted files, user attribution	Inventory table, evidence-linkage matrix, ILookIX workflow, limitations
3	Forensic Imaging and Safekeeping of a Suspect Storage Device	Acquisition, write protection, hashes, ASB container, carving, custody	Imaging flowchart, procedure, integrity explanation, custody form
4	Recovery of Deleted Evidence from Volume Shadow Copies	Evidence selection, Event Analysis, Cloud/Lead Analysis, VSS recovery	Classification table, hypothetical timeline, recovery workflow, validation
5	Multi-Source Investigation of a Suspected Cyber Incident	Windows, browser, email, messaging, mobile, internet/cloud evidence	Evidence map, cross-source matrix, challenge analysis, conclusion

Method and reporting principles

- Authority and scope: access only evidence covered by valid institutional, legal, or owner authorization.
- Preservation: protect originals, use write-blocking where appropriate, and create documented forensic copies.
- Integrity: calculate cryptographic hashes where technically possible and record tool versions and settings.
- Correlation: use independent artifacts to test a hypothesis rather than relying on one timestamp or one file.
- Transparency: distinguish observed facts, inferences, assumptions, and unresolved questions.
- Privacy: minimize collection of unrelated personal or institutional data and restrict access to case materials.

Scope note: all case descriptions, timelines, example artifacts and sample custody entries are illustrative. Actual investigations must use verified evidence, approved tools, accurate case records and the applicable institutional procedure.

Report organization: the five assignments move from server-access investigation and file-to-user linkage to forensic imaging, deleted-data recovery, and multi-source incident reconstruction. Each case study identifies the purpose, relevant evidence, examination approach, interpretation limits, and reporting considerations.

ASSIGNMENT – 2(1)

Suspected Unauthorized Access to a College Server

2.1 Case summary and objectives

A college administrator reports that confidential examination-related files were accessed and copied from a departmental server outside normal working hours. Potential evidence includes the administrator's complaint, a suspect workstation, server access logs, a recovered USB device, and relevant files. The objective is to determine what the available evidence supports while avoiding contamination of original evidence.

Working assumptions: the institution has authorized the investigation; the exact event time and user identity are initially unknown; clocks may differ between systems; and the USB device is only a potential source until examined. No conclusion of misconduct should be reached before corroboration.

Investigation objectives: preserve relevant logs and media, establish a reliable event timeline, test whether file access and copying occurred, identify the limits of account-level attribution, and report findings with supporting artifacts and alternative explanations.

Priority actions include preserving short-retention server and authentication logs, documenting the source and collection time, and comparing account activity with workstation and USB artifacts. Account ownership alone does not establish who performed an action; conclusions should be based on corroborated evidence and recorded alternatives.

2.2 Identification, Collection, Examination, Analysis and Presentation

Phase	Activities	Expected output / control
Identification	Record complaint, scope, affected server/share, likely time window, people with access, and potential devices/accounts.	Case ID, authority record, initial evidence map.
Collection	Photograph and document workstation/USB; export server and authentication logs using approved methods; record source, handler, time and tool.	Acquisition notes, evidence labels, custody entries.
Examination	Work from verified copies; inspect log events, file-system metadata, USB artifacts, relevant files and access records.	Reproducible artifact list with hashes and extraction details.
Analysis	Normalize time zones; correlate logins, file reads/copies, USB connection events and file timestamps; test alternate explanations.	Timeline, linkage reasoning, confidence and gaps.
Presentation	Explain methods, evidence, findings, limitations and conclusions in plain language; preserve technical appendices.	Final report, exhibits, evidence table and clear summary.

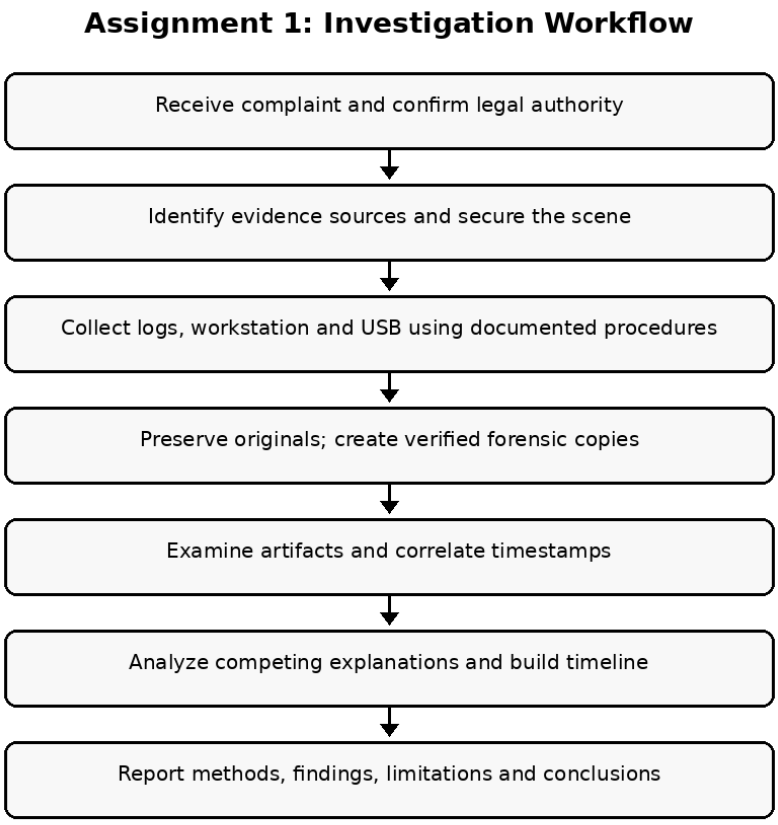


Figure 1. Proposed investigation sequence; the workflow is a plan, not a claim of completed forensic activity.

2.3 Potential evidence and investigative value

Evidence item	Potential value	How to examine / correlate	Caution
Administrator complaint	Defines allegation, scope and initial time window.	Record exact wording, date received and basis of knowledge.	An allegation is not proof of access.
Server access/security logs	May show account, source IP, authentication and file/share events.	Preserve original exports; inspect event IDs, log gaps and clock settings.	Logs may be incomplete, altered or use service accounts.
Suspect workstation	May contain login history, recent files, shortcuts, browser traces and USB artifacts.	Acquire image where authorized; inspect copies using validated tools.	Shared use or remote access may complicate attribution.
Recovered USB device	May contain copied files, file-system metadata and device identifiers.	Photograph, label, image with write protection and compare hashes.	Presence alone does not prove who copied a file.
Relevant examination files	Can confirm names, content, classification and similarity to server files.	Hash and compare copies; record paths and metadata.	A matching file may have been received from another source.
Authentication/network records	May corroborate account activity, IP assignment or remote connection.	Correlate with server and workstation events.	IP address does not necessarily identify a person.
Custody and access records	Show who handled evidence and whether access was controlled.	Review handovers, seals and audit records.	Missing documentation weakens defensibility.

2.4 Collection and preservation precautions

- Obtain written authority and define the systems, accounts, date range and data categories within scope.
- Photograph the scene and record device make/model, serial number, power state, connected peripherals and visible identifiers.
- Do not browse files casually on the original workstation or USB. If live collection is necessary, document each action and its possible effect.

- Use an appropriate write blocker when imaging storage media; use validated acquisition software and record version/settings.
- Maintain an evidence log with unique ID, description, collector, date/time, location, seal number and each transfer.
- Store originals in a secure, access-controlled location. Analyze working copies and verify their integrity.
- Preserve logs promptly because retention limits and log rotation may remove relevant records.

2.5 Constructing a timeline

A timeline should combine server access logs, authentication events, workstation logon records, USB connection artifacts and file metadata. Investigators should preserve the original timestamp and separately record the normalized timestamp, source time zone, clock offset (if known), and any uncertainty. Events should be ordered by time but not treated as causally linked solely because they are close together.

Hypothetical time (local)	Source / artifact	Possible interpretation	Confidence / check
21:02	Workstation logon event	A user session began.	Check account, logon type and clock.
21:11	Server authentication log	An account authenticated to the server.	Correlate source host/IP and authentication method.
21:14	File access audit event	A protected file may have been opened or read.	Confirm audit semantics and path.
21:19	USB connection artifact	A removable device was attached.	Correlate device ID and session; not proof of copying.
21:23	Matching file on USB image	A file resembling the protected document is present.	Compare cryptographic hashes and metadata.

The entries above are illustrative only and are not actual case findings. A defensible conclusion requires corroboration and consideration of normal administrative tasks, scheduled jobs, shared accounts and remote access.

2.6 Legal and ethical considerations

Investigators should follow institutional policy and applicable law, obtain proper authorization, preserve confidentiality of examination materials, and avoid accessing unrelated personal data. Evidence should be handled impartially; potentially exculpatory information must not be ignored. Access to evidence should be limited to authorized personnel, and reports should avoid naming a person as responsible unless the evidence supports that attribution.

2.7 Sample final forensic report structure

Report section	Content
Case details	Case identifier, requester, investigator, dates, authority and scope.
Question presented	What access or copying activity is being investigated.
Evidence received	Item IDs, descriptions, condition, source and custody history.
Methods	Acquisition method, tools/versions, settings, hashes and examination steps.
Findings	Artifacts observed, locations, relevant timestamps and supporting exhibits.
Analysis	How artifacts correlate, competing explanations and confidence level.
Limitations	Missing logs, clock uncertainty, shared accounts, unexamined sources.
Conclusion	Plain-language answer to the investigative question without overstating certainty.
Appendices	Evidence tables, timeline, hash list, custody records and tool output excerpts.

Illustrative server authentication and file-access log review – Assignment 2(1)



Note: illustrative interface mock-up created for presentation; it is not a screenshot from ILookIX, IXImager, or a live investigation. Replace with authentic, authorized screenshots if available.

ASSIGNMENT – 2(2)

Linking a Suspicious File to a User Using File-System Evidence

3.1 Case summary

A shared desktop contains a suspicious executable, hidden files, encrypted documents, browser-related artifacts and multiple user accounts. Investigators are asked to determine whether the artifacts can be linked to a particular account and to use a forensic suite such as ILookIX. File presence alone is insufficient to establish who created or used it.

3.2 Hardware and software inventory

Category	Details to document	Why it matters
Computer	Make/model, serial number, asset tag, condition, power state, connected devices.	Identifies the source and helps reproduce the examination.
Storage	Drive type, capacity, serial/unique ID, partition layout, attached USB media.	Establishes which media were examined and their relationships.
Operating system	Name, edition, build, installation details, time zone, system clock.	Supports interpretation of event and file timestamps.
File systems	NTFS/FAT/exFAT or other format, volumes, mount points.	Determines available metadata and recovery methods.
Accounts	Local/domain accounts, SIDs, profile paths, privilege levels.	Helps distinguish user profiles and shared/system activity.
Applications	Browser versions, security software, relevant utilities and forensic tools.	Explains artifact formats and possible automated activity.
Network context	Hostname, domain, IP configuration, mapped shares if relevant.	May link file activity to network resources.
Acquisition record	Image ID, method, date/time, hash, examiner and tool versions.	Supports integrity and repeatability.

3.3 How file-system evidence assists an investigation

Storage devices preserve file content and, depending on the file system and acquisition method, directory entries, allocation information, timestamps, access-control details and other metadata. Operating-system artifacts such as recent-file records, shortcut files, event logs, application history and user-profile paths may add context. These artifacts should be correlated rather than interpreted individually.

- File name and path: indicate where an artifact was found, but names can be changed and files can be copied.
- Creation/modification/access times: can help sequence activity, but copying, clock errors, software behavior and time-zone conversion can alter interpretation.
- User profile and account identifiers: may connect artifacts to a profile, but a profile does not prove the named person was operating the machine.
- Application and browser artifacts: may indicate downloads, execution attempts or navigation, subject to caching, synchronization and shared-device issues.
- File hashes: can identify identical content across locations; a hash match does not by itself identify the person who placed the file.

3.4 Hidden and encrypted files

Hidden attributes can reduce visibility during ordinary browsing but do not make a file invisible to forensic tools. Investigators should record the exact path, attributes, metadata and surrounding directory context. Encrypted files may conceal content unless authorized keys or credentials are available. Investigators should

preserve the encrypted original, document any lawful decryption attempt and avoid altering the evidence. They must not infer malicious intent merely because a file is hidden or encrypted.

3.5 Evidence-linkage matrix

Artifact / field	Example observation (hypothetical)	Linkage value	Alternative explanation / check
Executable path	C:\Users\UserB\Downloads\tool.exe	Places file within a profile path.	Profile may be shared; path can be copied or restored.
File metadata	Creation and modification times near incident window.	Helps sequence possible placement or changes.	Copy operations and clock differences may affect times.
Account artifacts	Recent-file record associated with UserB profile.	Supports activity within that profile.	Could be generated by preview/indexing or another person.
Execution evidence	Prefetch or application log references executable.	May support that a program ran.	Artifact availability depends on OS/version/settings.
Browser download record	Download URL and time align with file arrival.	Corroborates a possible acquisition route.	Browser profile may be shared or synchronized.
USB / external media	Matching hash appears on a USB image.	Shows identical content existed on another medium.	Does not establish direction or actor without more evidence.
Security logs	Logon/session aligns with relevant artifact times.	Adds account/session context.	Remote access, service accounts and log gaps must be assessed.

3.6 ILookIX-oriented analysis workflow

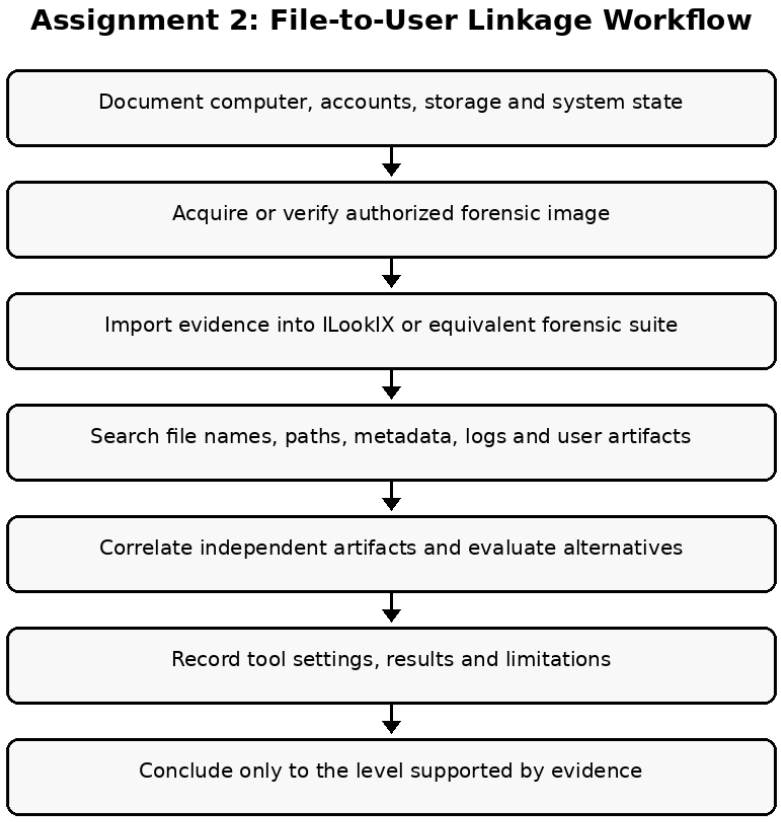


Figure 2. General workflow for file-to-user linkage. Actual menu names and supported formats depend on the installed ILookIX version.

1. Confirm authorization, case ID, source details and evidence integrity before importing.
2. Use the approved acquisition or verified forensic image; preserve the original evidence and work from a copy.

3. Create a case in ILookIX (or the approved suite), record the software version and case settings, and import the evidence using the supported workflow.
4. Review the file system, user profiles, hidden items, relevant metadata, logs and available browser/application artifacts.
5. Search for the suspicious filename, path, hash and related identifiers. Export relevant results with source path and context.
6. Correlate results across independent artifact types and document conflicting or missing evidence.
7. Save case notes and exports, validate important findings independently where possible, and record tool limitations.

3.7 Direct evidence versus contextual evidence

Type	Example	Interpretation
Potentially direct artifact	A trustworthy execution record associated with a specific account/session and the executable's hash.	Can strongly support execution in that session; still assess who controlled it.
Corroborating artifact	Browser download history matching file path/time.	Supports a possible route by which the file arrived.
Contextual artifact	A file exists in a user's Downloads folder.	Shows location, not necessarily creation or execution by the user.
Weak/ambiguous artifact	A generic filename or hidden attribute.	Insufficient alone to establish identity, intent or activity.

3.8 Limitations and conclusion

At least three alternative explanations must be considered: (1) multiple people used the same account or desktop; (2) the file was copied, restored, synchronized or placed by an automated process; (3) timestamps were altered or interpreted in the wrong time zone; (4) malware or remote access operated under another person's session; and (5) relevant logs are incomplete. Attribution should therefore be expressed as supported, not supported, or inconclusive, with the reasons stated.

Conclusion: file-system metadata and user artifacts can support a linkage hypothesis, but reliable attribution requires a consistent set of independent artifacts, integrity documentation and evaluation of alternatives.

Illustrative file-system and user-to-file linkage review – Assignment 2(2)



Note: illustrative interface mock-up created for presentation; it is not a screenshot from ILookIX, IXImager, or a live investigation. Replace with authentic, authorized screenshots if available.

ASSIGNMENT – 2(3)

Forensic Imaging and Safekeeping of a Suspect Storage Device

4.1 Collection and transportation

The case involves a seized laptop with an internal storage device and an attached USB drive. The aim is to preserve the original media, acquire a forensic image, verify integrity and maintain a documented chain of custody.

8. Confirm authority, define the items and record the purpose and scope of the seizure.
9. Photograph the laptop, screen, ports, connected USB, cables, labels and physical condition before changing the setup.
10. Record make/model, serial numbers, asset tags, date/time, power state, network connections and visible user/session information.
11. Do not casually power off or on the system. If it is running and volatile evidence may matter, follow the approved live-response procedure and document all actions; otherwise follow the validated seizure protocol.
12. Disconnect or isolate network access only as directed by the response plan, considering potential remote wiping and volatile evidence.
13. Label laptop, internal storage (if removed by a qualified examiner) and USB separately; package to prevent damage and unauthorized access.
14. Use tamper-evident seals and record each transfer, handler, time, purpose and seal condition.

4.2 Physical acquisition and why analysis uses an image

Physical acquisition attempts to capture the addressable contents of a storage medium, including allocated data and, where accessible, unallocated areas and remnants not visible through ordinary file browsing. A forensic image allows repeatable examination without repeatedly interacting with the original device. It also permits independent verification and protects the source from accidental writes. The achievable scope depends on device type, encryption, bad sectors, controller behavior and acquisition method.

A sound acquisition record should identify the source device, examiner, date and time, acquisition tool and version, selected acquisition mode, destination media, errors encountered and verification results. If a device is encrypted or contains unreadable sectors, the report should state the limitation and describe how it affects the scope of the image. The examiner should preserve the acquisition log and avoid presenting a partial or logical acquisition as a complete physical image. The examiner should also confirm that source and destination identifiers are correct, record any read errors, and verify hashes over the same data scope whenever the acquisition method supports that comparison. The original media, acquired image and acquisition log should remain access-controlled and available for independent review.

4.3 IXImager imaging workflow

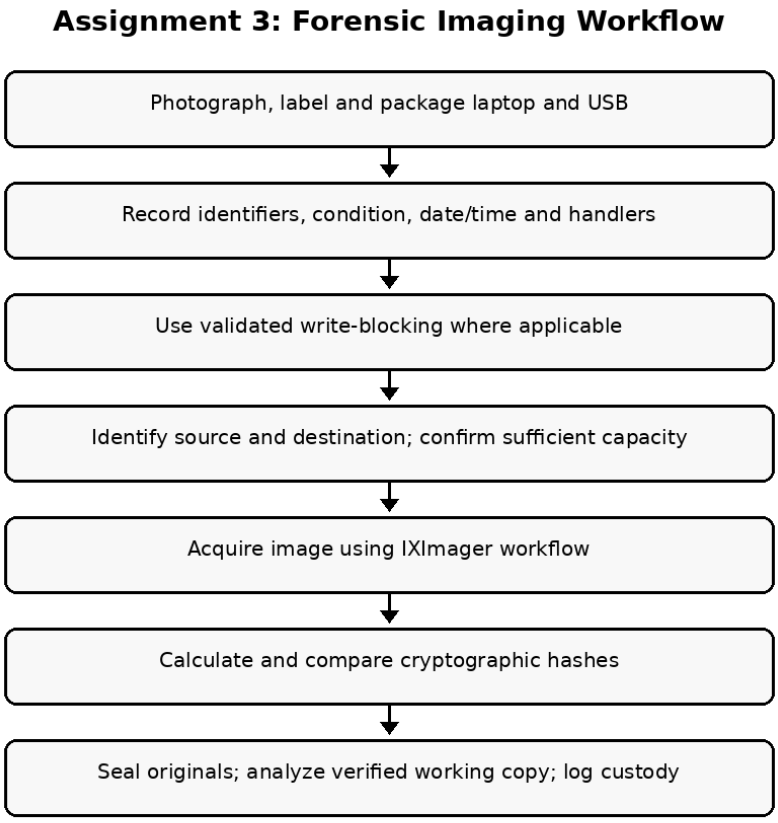


Figure 3. Planned imaging and safekeeping workflow. The exact IXImager options should be checked against the version and laboratory procedure.

Step	Procedure	Documentation / verification
1. Source identification	Confirm the correct laptop drive and USB using labels, model, serial/unique identifiers and capacity.	Photographs, identifiers, source ID and examiner.
2. Write protection	Attach a suitable validated hardware write blocker where applicable; verify expected operation.	Blocker make/model, status and any validation record.
3. Destination check	Choose a clean evidence destination with sufficient free capacity; identify the output format and case path.	Destination ID, available capacity and output settings.
4. Acquisition	Use IXImager according to the approved procedure; select the source carefully and begin acquisition.	Tool/version, date/time, acquisition mode, errors and progress log.
5. Verification	Enable or perform supported verification; calculate cryptographic hash values for image/source as appropriate.	Algorithm, values, verification result and any read errors.
6. Evidence sealing	Secure original devices and preserve the image as a controlled evidence item.	Seal number, storage location, access controls and custody log.
7. Analysis	Open a verified working copy in the forensic suite, not the original source.	Working-copy ID, hash check and analyst notes.

4.4 Hash values and integrity

A cryptographic hash such as SHA-256 produces a fixed-length digest for a particular byte sequence. If the source and acquired data are hashed over the same scope using the same algorithm, matching values support that the data are identical for that scope. Hashes do not prove that the source was authentic before collection, that the correct device was selected, or that no data were omitted because of unreadable sectors. Any acquisition errors and verification limits must be recorded.

Example evidence item	Algorithm	Illustrative hash / status	Meaning
USB image	SHA-256	Example only: 9f2a… (placeholder)	Replace with actual computed value; never present placeholder as a real result.
Laptop image	SHA-256	Example only: 71c4… (placeholder)	Record full digest and the precise data scope.
Working copy	SHA-256	Compare with verified image copy	A match supports copy integrity.

4.5 ASB container and accompanying information

The assignment brief names the ASB container format. Its exact capabilities should be confirmed against the relevant software documentation and syllabus; this report does not assume unsupported features. A forensic evidence container or package should be accompanied by case ID, item ID, source description, acquisition date/time, examiner, tool/version, acquisition settings, hash algorithm and value, verification status, error log, notes, and any password/key handling restrictions. The package must be stored securely and access logged.

4.6 Data carving

Data carving searches raw data for recognizable file signatures or structures, often in unallocated space or where directory/file-system metadata is missing or damaged. It can recover fragments or complete files without relying on a valid directory entry. Carving may lose filenames, paths, ownership, timestamps and directory relationships; recovered content may be incomplete or falsely identified by a signature. Carved results must be validated and described with these limitations.

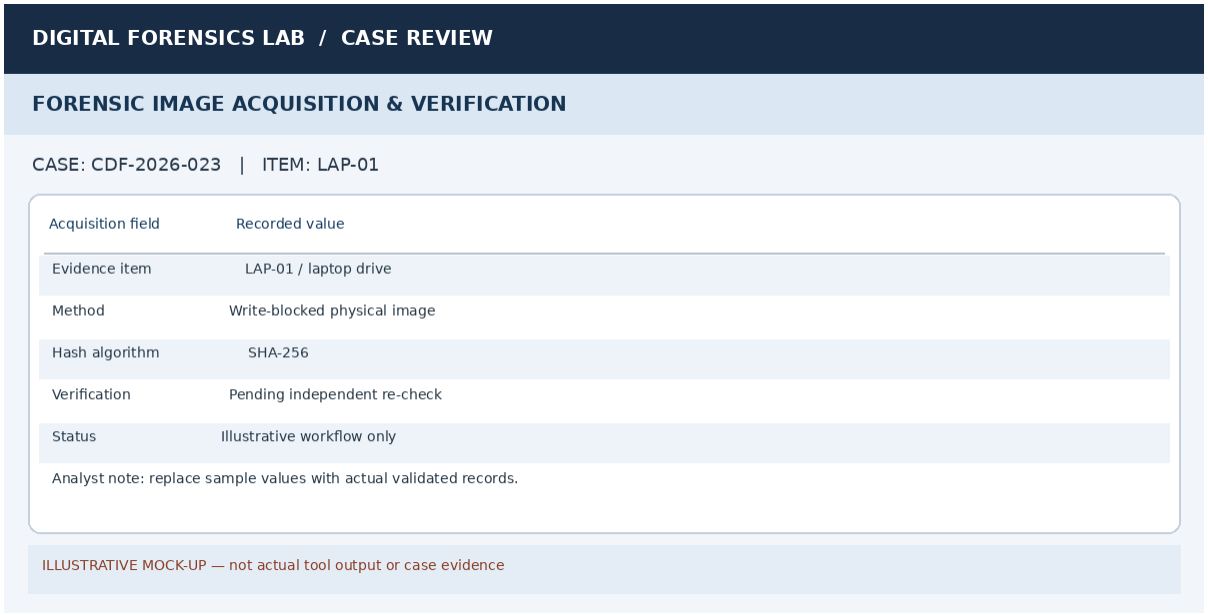
4.7 Time and chain of custody

Digital timestamps may reflect UTC, local time, device time, server time or application-specific conventions. Daylight-saving settings, incorrect clocks, copying, restoration and metadata semantics can complicate interpretation. Record raw timestamps, the source time zone, normalized time and the method used to convert them. Do not silently adjust or discard conflicting values.

Date/time	Item ID	Action / purpose	Released by	Received by	Seal / storage / notes
15-Oct-2026, 09:10	LAP-01	Collected and photographed	Examiner A	Examiner A	Seal S001; initial collection
15-Oct-2026, 09:35	USB-01	Packaged separately	Examiner A	Evidence custodian	Seal S002; USB identified
15-Oct-2026, 11:00	LAP-01	Transferred to lab intake	Evidence custodian	Examiner B	Seal checked intact
15-Oct-2026, 13:30	IMG-01	Image created and verified	Examiner B	Evidence custodian	Hash and tool log attached

The custody entries above are a sample form for the report and do not represent actual handling. Replace them with the true dates, times, names, item IDs and results if used for a real case.

Illustrative forensic acquisition and hash-verification workflow – Assignment 2(3)



Note: illustrative interface mock-up created for presentation; it is not a screenshot from ILookIX, IXImager, or a live investigation. Replace with authentic, authorized screenshots if available.

ASSIGNMENT – 2(4)

Recovery of Deleted Evidence from Volume Shadow Copies

5.1 Case summary and evidence selection

An employee is suspected of deleting documents and browser-related artifacts after an unauthorized activity. A forensic image of the Windows system is available, and Volume Shadow Copies (VSS) are present. Investigators must identify relevant evidence, compare earlier versions, validate recovered artifacts and reconstruct events without assuming deletion proves wrongdoing.

5.2 Evidence classification scheme

Class	Examples	Handling / analytical purpose
User-created files	Documents, spreadsheets, downloads, notes, user-created archives.	Compare content, metadata, paths and versions; determine relevance.
System/application files	Windows logs, browser databases, application configuration, caches.	Explain system state and activity; distinguish routine system behavior.
Potentially suspicious files	Unusual executable, renamed archive, files near incident period.	Examine hashes, signatures, provenance and execution artifacts.
Deleted/recovered artifacts	Deleted directory entries, carved files, VSS versions.	Record recovery method, completeness and metadata limitations.
Supporting/contextual evidence	Authentication logs, email, USB artifacts, backup/catalog records.	Corroborate timeline and evaluate alternative explanations.
Irrelevant/out-of-scope data	Unrelated personal files or unrelated work documents.	Minimize access and document exclusion according to scope/policy.

5.3 Event Analysis, Cloud Analysis and Lead Analysis

Event Analysis organizes artifacts into a time-ordered sequence to test whether actions are consistent with a hypothesis. It can reveal a possible relationship between logon, file access, browser activity, copying and deletion, but proximity in time is not proof of causation.

Cloud Analysis examines authorized cloud-related records such as synchronization logs, account access records, cloud storage metadata and provider audit logs. It may help determine whether a file was uploaded, downloaded, synchronized or accessed from another device. Lead Analysis follows investigative leads generated by artifacts—for example, a username, external device ID, URL or file hash—to identify further sources that may confirm or refute the current hypothesis. Each lead should be documented and independently validated.

5.4 Volume Shadow Copy analysis workflow

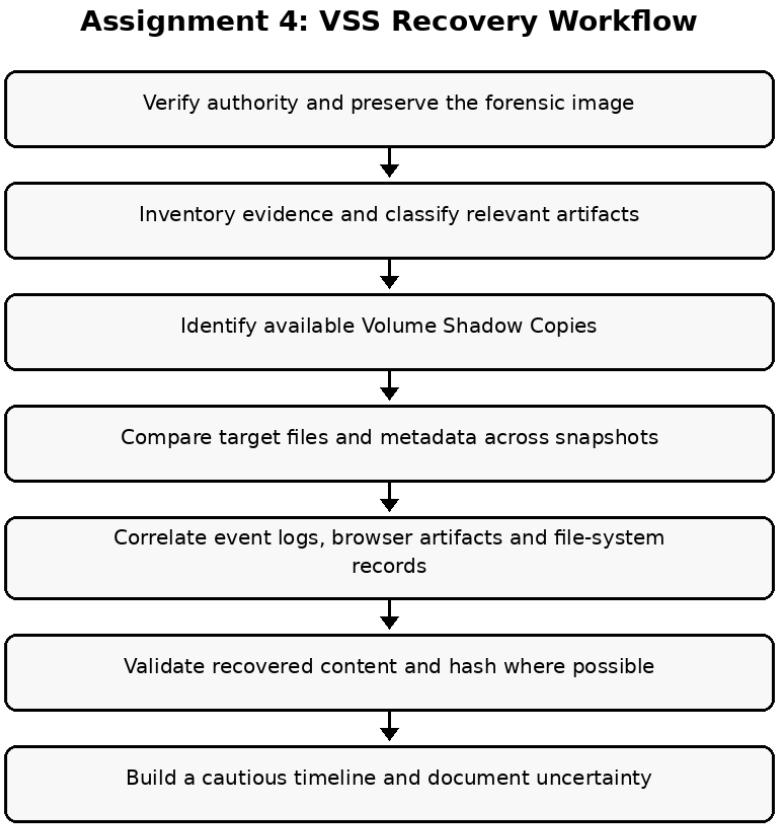


Figure 4. Proposed Volume Shadow Copy recovery workflow.

15. Verify the forensic image's integrity and record the Windows version and relevant volumes.
16. Inventory VSS snapshots and note snapshot identifiers, creation times and available contents.
17. Locate the target paths or artifacts in relevant snapshots; export copies through the forensic tool's supported method.
18. Compare current and historical versions using content, hashes, metadata and surrounding directory context.
19. Correlate changes with event logs, browser records, recent-file artifacts and other evidence.
20. Preserve recovered outputs with source snapshot ID, original path, extraction method, hash and examiner notes.
21. Validate important results and document if only fragments or metadata—not complete content—were recovered.

5.5 Hypothetical timeline

Time (illustrative)	Event	Possible evidence	Interpretive note
09:05	Document created	File metadata in current volume or snapshot	Creation time may not equal first authoring time.
09:18	Document modified	Earlier/later VSS version; file metadata	Compare contents and metadata; identify clock basis.
09:26	Possible copy to removable media	USB connection artifact and matching file	A connection alone does not prove a copy.
09:42	Document deleted	Current file-system record or event evidence	Deletion may be routine; establish context.

10:00	VSS snapshot available	Snapshot catalog and earlier version	Snapshot time does not necessarily equal file modification time.
Later examination	Earlier version recovered	Exported VSS artifact and hash	Record recovery source and completeness.

This timeline is hypothetical and demonstrates how to organize evidence; it is not a finding about an actual employee.

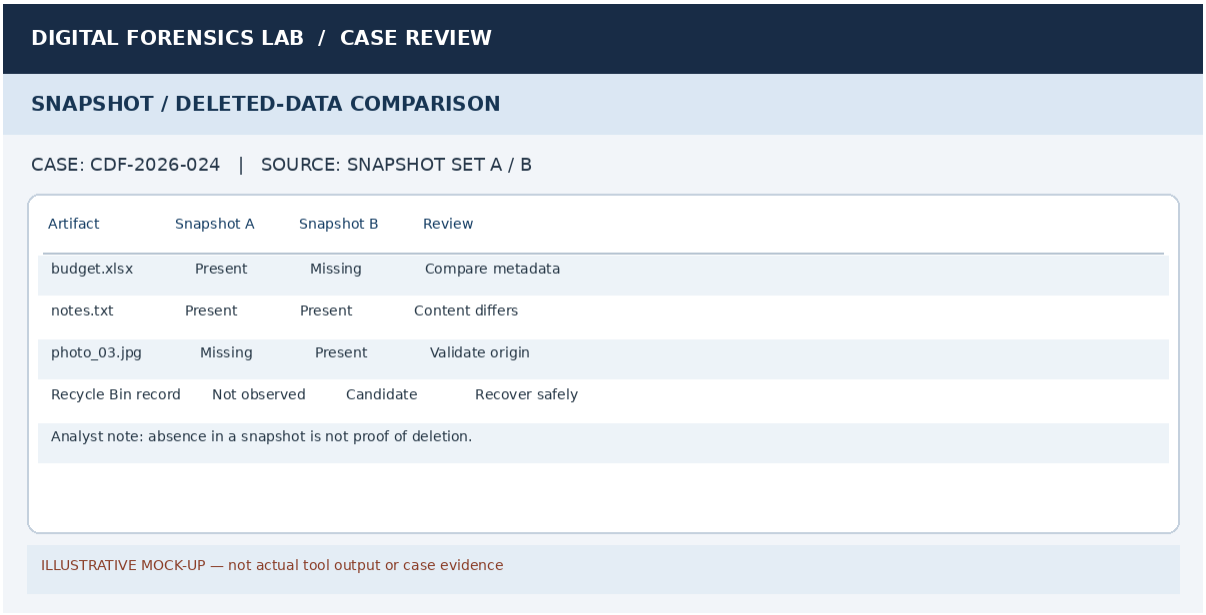
5.6 Validation checklist

- Confirm the source image hash and snapshot identifier.
- Record the original path, snapshot time, extraction method and tool/version.
- Calculate a hash for the recovered output where possible and preserve the original export.
- Compare content across snapshots and distinguish exact matches from partial or reconstructed data.
- Cross-check the artifact against an independent source, such as a log, browser record or backup.
- Assess time-zone, clock drift, file-system semantics and possible copying/restoration effects.
- Record errors, gaps, competing explanations and the reason the artifact is relevant.

5.7 Error sources and conclusion

Possible errors include misinterpreting local time as UTC, system clock drift, timestamp changes caused by copying/restoration, incomplete snapshots, overwritten or truncated content, false-positive carving, tool parsing differences, and confusing the snapshot creation time with the event time. Investigators should not infer intent from deletion alone. A defensible report explains what the recovered artifact shows, how it was validated and what it cannot establish.

Illustrative snapshot and deleted-data comparison – Assignment 2(4)



Note: illustrative interface mock-up created for presentation; it is not a screenshot from ILookIX, IXImager, or a live investigation. Replace with authentic, authorized screenshots if available.

ASSIGNMENT – 2(5)

Multi-Source Investigation of a Suspected Cyber Incident

6.1 Case summary and evidence map

A student account is suspected of being used in unauthorized access to an online service. With proper authorization, investigators may examine a Windows computer, browser artifacts, email, messaging records, a mobile phone and relevant internet/cloud records. The central task is to correlate sources while accounting for differences in clocks, retention, ownership and provider access.

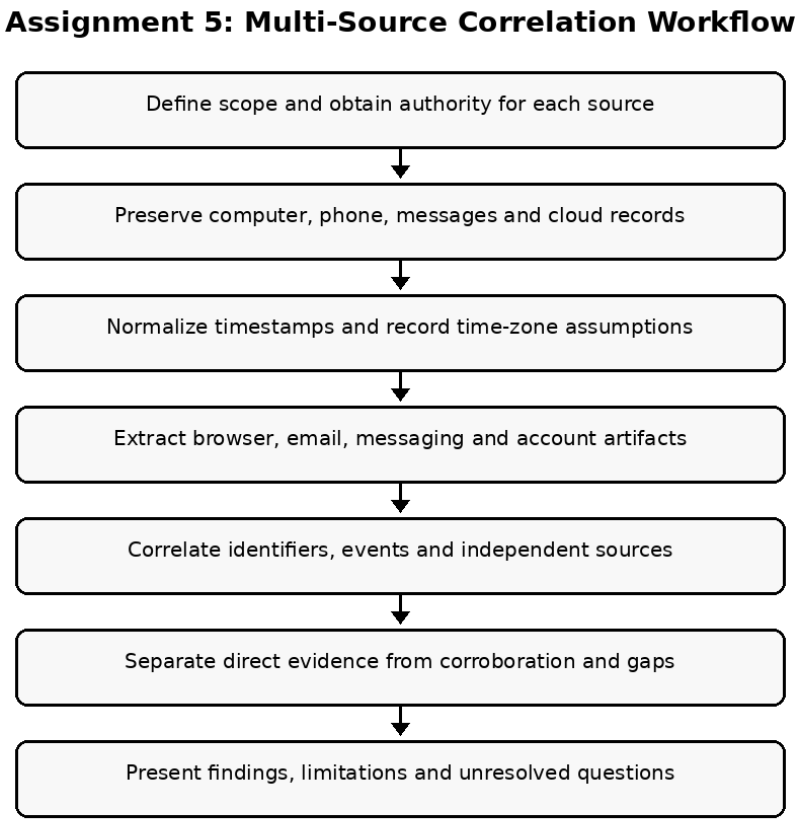


Figure 5. Multi-source evidence correlation workflow.

Source	Potential artifacts	Possible investigative question
Windows computer	Logon events, user profiles, file-system metadata, execution traces, network configuration.	Was the device active during the relevant period?
Browser	History, downloads, cookies/session data where available, cache, saved form or site records.	Was a service visited or a file downloaded?
Email	Headers, sender/recipient, timestamps, message IDs, attachments, provider login records.	Was a link or attachment received or sent?
Messaging	Message records, attachment metadata, participant identifiers, timestamps.	Did communications coordinate or explain an event?
Mobile phone	App artifacts, device identifiers, notifications, local records, backup data.	Was the account or communication also used on mobile?
Internet/cloud	Provider audit logs, account sign-ins, IP/session records, cloud file metadata.	Was there a remote session or synchronization event?

6.2 Windows and other operating-system evidence

Windows may provide event logs, user-profile data, file-system metadata, application execution traces, browser artifacts and device connection records. Other operating systems may provide their own authentication logs, shell history, application databases, file metadata and system logs. The exact artifacts depend on the operating-system version, settings, retention and acquisition method. Investigators should document what was available and avoid assuming every artifact exists on every system.

6.3 Browser artifacts and interpretation

Browser history can show recorded visits; download records may identify a URL and destination; cache may preserve content; cookies or session databases may show state; and saved form or autofill data may contain user information. These artifacts are sensitive and must be collected within scope. Browser artifacts do not always prove a person intentionally visited a site: prefetching, embedded content, shared profiles, synchronization, extensions, automated processes and remote control can produce misleading traces. A URL record alone does not prove that a transaction was completed.

6.4 Email and messaging evidence

Email evidence may include full headers, sender and recipient fields, message IDs, server routing, timestamps, attachment hashes and provider access logs. Messaging evidence may include conversation records, participant identifiers, delivery/read status, attachment references and local database artifacts. Correlation should consider message forwarding, spoofed display names, account compromise, delayed delivery, unsynchronized clocks and missing messages. Preserve the original exported format and document how provider or device data were obtained.

6.5 Mobile forensic considerations

- Record device make/model, serial or IMEI where lawfully available, OS/version, lock state, SIM/eSIM and network state.
- Obtain valid authority and choose a method appropriate to the device and investigative question.
- Consider whether isolation may prevent remote wipe or, conversely, affect volatile or cloud-synchronized evidence; follow laboratory policy.
- Preserve the device condition and document any unlocking, cable connection, software interaction or power-state change.
- Understand that encryption, secure hardware, app sandboxing, operating-system version, deleted data and cloud dependence can limit extraction.
- Do not bypass access controls outside legal authority; protect unrelated personal and third-party data.

6.6 Internet and cloud evidence

Potential sources include provider sign-in history, IP/session logs, cloud file version history, sharing permissions, synchronization logs, DNS or proxy records, firewall logs and web-service audit trails. Collection may require provider preservation requests, account-holder consent, institutional authority or legal process. Records may be retained for limited periods, use provider-specific timestamps and omit content. Investigators should record the source, export method, time zone, account identifier and any gaps.

6.7 Cross-source correlation matrix

Hypothetical artifact	Source	Correlated artifact	What it may support	Caution
21:11 account sign-in	Cloud service	21:10 workstation logon	Device/session activity may align with account access.	Shared account, VPN or remote session possible.
21:14 service URL	Browser history	Provider audit record near 21:14	A browser session may relate to the service event.	History may be automatic or profile-shared.
21:16 email with link	Email system	Browser URL/download record	Possible link between communication and access.	Message receipt does not prove the link was opened.
21:20 attachment hash	Messaging app	Same hash in local downloads	Identical content may exist across sources.	Hash match proves content identity, not sender intent.
21:25 mobile notification	Mobile device	Account session alert from provider	May corroborate that an account event occurred.	Notification time and device ownership need validation.
21:30 cloud file version	Cloud storage	Local file metadata/hash	May indicate synchronization or file transfer.	Sync clients can act automatically.

6.8 Major challenges in multi-source forensics

Challenge	Impact	Mitigation
Different time zones / clock drift	Events may appear out of order.	Preserve raw times, document offsets and normalize transparently.
Encryption and locked devices	Content may be unavailable.	Use lawful, validated methods; document inaccessible evidence.
Cloud-provider limits	Logs may be incomplete, delayed or unavailable.	Request preservation promptly and record provider scope/retention.
Different data formats	Artifacts may not align automatically.	Maintain source metadata and validate parsers/exports.
Shared or compromised accounts	Account activity may not identify a person.	Correlate device, session, network and independent records.
Volatile / short retention	Evidence may disappear quickly.	Prioritize lawful preservation and document collection timing.
Privacy and jurisdiction	Some data may be outside authorized scope or location.	Confirm authority and minimize unrelated collection.
Tool limitations	Parser errors or unsupported artifacts may mislead.	Record versions, validate critical outputs and retain originals.

6.9 Final investigative conclusion

A final conclusion should separate three categories. Direct evidence consists of artifacts that directly record an event or content, subject to authenticity and context checks. Corroborating evidence consists of independent artifacts that are consistent with the event and strengthen the interpretation. Unresolved questions include identity of the actual operator, missing provider logs, ambiguous timestamps, shared-device use and whether a recorded session completed the alleged action.

Finding category	Illustrative example	Reporting language
Direct evidence	A validated provider audit record records a successful session for an account.	“The provider record indicates a successful session for this account at the recorded time.”
Corroborating evidence	Browser and workstation artifacts align with the provider event.	“These artifacts are consistent with activity on the examined workstation.”
Unresolved question	No reliable artifact identifies the individual at the keyboard.	“The available evidence does not independently establish who operated the session.”

Overall, multi-source correlation can strengthen a reconstruction, but attribution must remain proportionate to the quality and independence of the artifacts. Contradictory or missing evidence should be stated rather than hidden.

Illustrative multi-source incident timeline and correlation – Assignment 2(5)



Note: illustrative interface mock-up created for presentation; it is not a screenshot from ILookIX, IXImager, or a live investigation. Replace with authentic, authorized screenshots if available.

7. OVERALL CONCLUSION

The five case studies demonstrate that digital forensics depends on systematic identification, collection, preservation, examination, analysis and presentation. Evidence integrity and chain of custody are essential because the usefulness of a technical artifact depends not only on its content but also on how it was obtained, stored and interpreted.

Server logs, file-system metadata, forensic images, Volume Shadow Copies, browser records, mobile artifacts and cloud logs can each contribute to an investigation. No single artifact should ordinarily be treated as conclusive when plausible alternatives remain. Investigators should correlate independent sources, document timestamps and tool behavior, preserve originals, and communicate the difference between observation and inference.

The most defensible report is transparent about scope, methods, hashes, evidence gaps, limitations and uncertainty. In an academic or institutional setting, legal authority, confidentiality, proportionality and respect for unrelated personal information are as important as technical skill.

8. REFERENCES AND ASSUMPTIONS

Source document

CR23532 – Cyber and Digital Forensics, Case Study Assignments-II, Rajalakshmi Engineering College, six-page assignment brief supplied for this report. The assignment titles, case backgrounds, question coverage and expected deliverables in this report are based on that brief.

Technical assumptions and cautions

- All timelines, example artifacts, hash snippets and chain-of-custody entries are hypothetical examples for educational illustration; they are not actual forensic results.

- Use the institution’s prescribed syllabus and current tool documentation for the exact operation of ILookIX, IXImager and the ASB container format. This report does not claim to have executed these tools.
- Use current applicable laws, institutional policies and laboratory standard operating procedures for authorization, privacy, evidence seizure, storage and disclosure.
- Where a specific tool or artifact is unavailable, record the limitation and use an approved alternative only if permitted by the case scope.